

PORTLAND INTL JETPORT, ME, USA

WMO: 726060

Lat: 43.650N

Lon: 70.317W

Elev: 45

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	5.0	-13.9	2.6	3.1	-9.2	3.4	7.7	27.1	27.3	24.3	24.9	6.9	280	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	86.6	71.6	83.3	69.9	80.4	68.2	74.1	83.1	72.3	80.0	70.6	77.2	10.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.1	115.0	78.7	69.6	109.1	76.3	68.2	103.6	74.7	37.6	83.2	36.0	80.3	34.5	77.1	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	19.5	17.3	DB	-8.3	92.5	5.6	3.1	-12.4	94.7	-15.7	96.5	-18.8	98.3	-22.9	100.5	(4)
(5)			WB	-9.0	77.2	5.5	1.8	-12.9	78.5	-16.2	79.5	-19.2	80.5	-23.2	81.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	23.5	25.9	33.7	44.4	54.3	63.6	69.8	68.6	61.4	50.2	39.7	30.2
	DBStd	17.67	10.49	9.45	8.61	6.94	6.62	6.56	5.03	4.89	6.60	7.17	8.09	8.93
	HDD50	3254	823	675	510	195	26	0	0	0	2	87	321	615
	HDD65	6891	1287	1095	971	619	342	102	12	19	145	461	759	1079
	CDD50	2240	0	0	4	26	158	409	614	578	345	93	12	1
	CDD65	402	0	0	0	0	9	61	161	132	37	2	0	0
	CDH74	2886	0	0	4	15	124	491	1143	852	239	18	0	0
	CDH80	729	0	0	1	3	35	132	302	197	57	2	0	0
Wind	WSAvg	7.5	7.9	8.4	9.1	8.9	7.8	6.9	6.5	6.2	6.4	7.3	7.5	7.6
Precipitation	PrecAvg	44.9	3.6	3.3	3.8	4.1	3.6	3.3	3.1	3.0	3.4	4.3	4.8	4.4
	PrecMax	66.4	11.9	7.1	9.3	10.0	9.2	9.0	7.8	15.2	8.8	14.8	13.5	10.0
	PrecMin	29.0	0.3	0.0	0.8	0.3	0.5	0.7	0.6	0.5	0.6	0.6	0.9	1.2
	PrecStd	8.5	2.1	1.6	1.8	2.0	1.9	1.7	1.5	2.3	1.6	3.2	2.5	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	51.8	66.1	75.0	84.9	89.0	91.5	89.6	86.8	76.6	64.1	56.4
		MCWB	49.5	45.1	53.6	58.5	65.9	72.3	74.1	73.5	72.4	63.4	55.5	50.0
	2%	DB	47.1	46.4	54.4	65.2	76.8	83.9	86.4	84.9	80.1	69.4	59.2	50.7
		MCWB	43.6	41.4	45.3	51.6	62.5	69.4	71.5	71.4	67.9	60.1	53.7	47.0
	5%	DB	42.5	42.9	49.1	60.2	71.3	79.6	83.2	81.6	75.9	64.9	55.9	47.3
		MCWB	39.0	38.3	42.4	49.1	59.4	66.4	70.5	69.6	65.8	58.1	51.4	43.8
	10%	DB	38.2	39.5	45.3	56.0	66.3	75.6	80.2	78.8	72.3	61.5	52.6	43.3
		MCWB	34.5	35.5	39.8	46.8	56.4	64.8	68.8	67.8	64.2	55.8	48.6	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	47.0	54.8	59.9	68.7	74.9	76.9	76.2	73.4	66.5	58.9	52.3
		MCDB	51.1	49.9	62.4	71.0	79.1	85.9	87.6	85.7	83.2	71.4	62.3	54.7
	2%	WB	44.3	42.7	48.0	55.0	64.9	71.8	74.3	73.8	70.7	62.5	55.6	48.1
		MCDB	46.0	45.6	52.8	61.7	73.5	79.8	83.1	81.9	77.0	67.0	57.6	50.0
	5%	WB	39.3	39.0	43.7	51.1	61.3	69.3	72.5	71.8	68.4	59.8	52.3	44.2
		MCDB	42.1	41.8	48.0	57.0	69.6	76.4	80.2	78.5	73.2	63.9	54.8	46.6
	10%	WB	34.8	35.9	40.3	48.1	57.9	66.8	70.6	69.9	66.1	57.1	49.0	39.9
		MCDB	37.6	39.2	44.3	54.8	64.7	73.3	77.5	75.5	70.5	60.7	52.3	42.9
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.6	15.3	17.3	17.1	17.1	17.1	17.6	17.7	17.2	15.6	14.9
		MCDBR	18.1	18.2	20.3	24.6	26.0	23.0	21.1	21.3	21.7	21.6	18.4	18.9
		MCWBR	16.6	14.7	14.5	15.0	14.3	12.0	9.9	10.6	12.2	14.1	14.6	16.4
	5% WB	MCDBR	18.1	17.3	18.6	20.9	23.1	20.3	18.2	18.2	17.9	18.1	15.7	18.1
		MCWBR	17.4	15.3	14.7	14.9	14.1	12.1	9.7	10.2	11.5	13.7	14.3	17.4
Clear-Sky Solar Irradiance	taub	0.291	0.304	0.329	0.356	0.390	0.442	0.456	0.423	0.379	0.338	0.319	0.295	
	taud	2.451	2.421	2.434	2.416	2.366	2.241	2.216	2.299	2.400	2.510	2.526	2.514	
	Ebn at Noon	268	284	289	289	280	264	259	264	268	268	256	257	
	Edh at Noon	23	28	32	35	38	43	44	39	33	26	22	20	
All-Sky Solar Radiation	RadAvg	530	810	1166	1463	1637	1757	1845	1679	1350	869	566	423	
	RadStd	37	61	71	155	189	144	126	87	77	78	52	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+54
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	N/A

Nomenclature: See separate page